

Accessing an EC2 Instance in a Private Subnet

To Begin with the Lab

Summary of the Lab

This lab demonstrated secure access to an EC2 instance deployed in a private subnet using two methods: an EC2 Instance Connect Endpoint and a Bastion Host in a public subnet. A Bastion Host with a public IP was used to SSH into the private Database Server through its private IP, while the EC2 Instance Connect Endpoint provided browser-based secure access using AWS authentication. Since the private instance initially had no internet connectivity, a NAT Gateway was created in the public subnet with an Elastic IP, and the private route table was updated to route outbound traffic through it. After the configuration, the private EC2 instance successfully accessed the internet while remaining protected from direct inbound internet connections, ensuring a secure and highly available network architecture.

- **Note:** This lab assumes that the VPC, public and private subnets, Internet Gateway, and Route Tables have already been configured in the previous lab.
- Open the **AWS Management Console** and navigate to the **EC2** service. Launch a new EC2 instance in **Public-Subnet-1**, assign it a meaningful name such as **Bastion-Host**, enable **Auto-assign Public IP**, and use an appropriate security group that allows SSH access.
- Also Select the AMI as Amazon Linux

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name
 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **My AMIs** **Quick Start**

[Amazon Linux](#) [macOS](#) [Ubuntu](#) [Windows](#) [Red Hat](#) [SUSE Linux](#) [Debian](#)

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-06067086cf86c58e6 (64-bit (x86), uefi-preferred) / ami-071bc399fae13263b (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Summary

Number of instances Info

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-06067086cf86c58e6

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

- Instance type as t3 micro and use the existing key, otherwise create a new key-pair if you don't have one.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

jenkins

[Create new key pair](#)

- Choose the VPC and Use **Public-Subnet-1** and enable **Auto-assign Public IP**, also name the security group

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-02a0440be172ac2c9 (myfirst-vpc)

192.168.0.0/24

Subnet [Info](#)

subnet-07dba1f436052d9b5

Public-Subnet-1

VPC: vpc-02a0440be172ac2c9 Owner: 463646775279
Availability Zone: us-east-1a (use1-az4) Zone type: Availability Zone
IP addresses available: 59 CIDR: 192.168.0.0/26

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

web-server

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@,.[*+=&(){}\$*

Description - required [Info](#)

launch-wizard-34 created 2026-06-30T06:27:35.588Z

Inbound Security Group Rules

► Security group rule 1 (TCP 22 0.0.0.0/0)

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12....[read more](#)

ami-06067086cf86c58e6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Preview code](#)

- Click Launch Instance

EC2 > Instances > Launch an instance

Launching instance

Launch initiation

80%

Details

- Wait until the Bastion Host is in the Running state and note its Public IP Address. This instance will act as the entry point for accessing resources located inside the private subnet.
- You have to run this command to give the full access to the key that you have created

```
chmod 400 "your_key_name"
```

- Let's connect to the instance

```
manr@LAPTOP-70U805T1 MINGW64 ~
$ cd Downloads/

manr@LAPTOP-70U805T1 MINGW64 ~/Downloads
$ chmod 400 jenkins.pem
$ ssh -i jenkins.pem ec2-user@3.80.85.121

_ _ _ _ _
##      Amazon Linux 2023
##
##      https://aws.amazon.com/linux/amazon-linux-2023
##
_ _ _ _ _

[ec2-user@ip-192-168-0-60 ~]$
```

- Launch another EC2 instance and place it inside Private-Subnet-1. Name it Database-Server, disable Auto-assign Public IP, and create the instance so that it remains isolated from direct internet access.

Network settings Info

VPC - required Info
vpc-02a0440be172ac2c9 (myfirst-vpc-192.168.0.0/24)

Subnet Info
subnet-0d66e2276cf7cdf66 Private-Subnet-1
VPC: vpc-02a0440be172ac2c9 Owner: 463646775279
Availability Zone: us-east-1a (use1-az4) Zone type: Availability Zone
IP addresses available: 59 CIDR: 192.168.0.128/26

Auto-assign public IP Info
Disable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☒ Create security group ☐ Select existing security group

Security group name - required
database-server

Description - required Info
launch-wizard-34 created 2026-06-30T06:36:45.705Z

Inbound Security Group Rules
▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
ami-06067086cf86c58e6

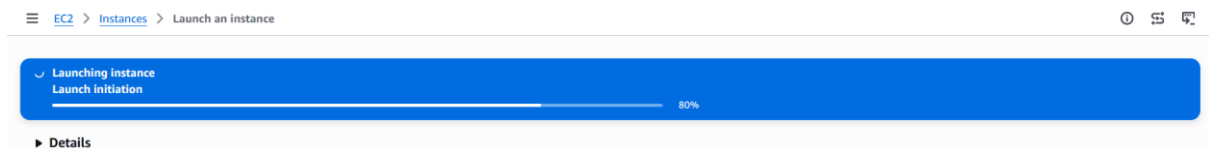
Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

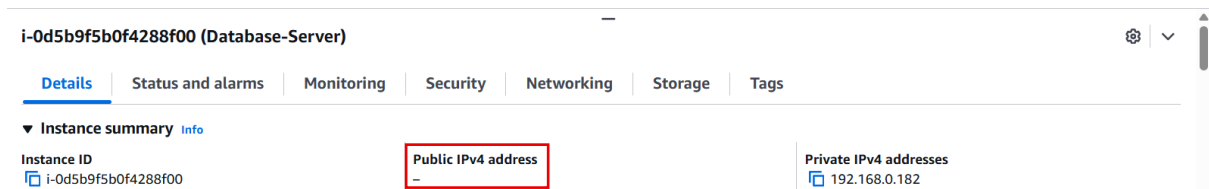
Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

- Click Launch Instance



- Verify that the Database Server has only a **Private IP Address** and no Public IP Address assigned. Since the instance resides in a private subnet, it cannot be accessed directly from your local machine using SSH.



Access Using EC2 Instance Connect Endpoint

Note: Understand that **EC2 Instance Connect Endpoint** requires authentication through your AWS account. Only users with valid AWS credentials and the required IAM permissions can use this method to connect to private EC2 instances.

- Open the **VPC** service and navigate to the **Endpoints** section.
- Click **Create Endpoint**
- Give the Name and choose the **EC2 Instance Connect Endpoint** service

Create endpoint [Info](#)
Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings
Specify a name and select the type of endpoint.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

Private-Subnet-EC2-Endpoint

Type [Info](#)
Select a category

☐ AWS services
Connect to services provided by Amazon with an interface endpoint, or a Gateway endpoint.

☐ PrivateLink Ready partner services
Connect to SaaS services which have AWS Service Ready designation with an interface endpoint. Uses AWS PrivateLink.

☐ AWS Marketplace services
Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint.

☒ **EC2 Instance Connect Endpoint**
An elastic network interface that allows you to connect to resources in a private subnet.

☐ Resources
Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink.

☐ Service networks
Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink.

☐ Endpoint services that use NLBs and GWLBs
Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint.

select your existing VPC and choose the default security group

Network settings
Select the VPC in which to create the endpoint.

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-02a0440be172ac2c9 (myfirst-vpc)

► **Additional settings**

Security groups (1/5) [Info](#)

Q Search

VPC ID : vpc-02a0440be172ac2c9 X Clear filters

Group ID	Group name	VPC ID	Description
☐ sg-078ddde12454c44eb	web-server	vpc-02a0440be172ac2c9	launch-wizard-34 created 2026-06-30T06:27:35.588Z
☒ sg-0844122bf83c2111c	default	vpc-02a0440be172ac2c9	default VPC security group
☐ sg-0c2d5464fe740f4ad	launch-wizard-31	vpc-02a0440be172ac2c9	launch-wizard-31 created 2026-06-29T10:55:26.944Z
☐ sg-0bd0d61866ceb1e10	launch-wizard-32	vpc-02a0440be172ac2c9	launch-wizard-32 created 2026-06-29T11:07:44.842Z
☐ sg-0cf303ca55b9e3e42	database-server	vpc-02a0440be172ac2c9	launch-wizard-34 created 2026-06-30T06:36:45.705Z

sg-0844122bf83c2111c X

choose **Private-Subnet-1**, select the appropriate Security Group, and create the endpoint.

Subnet
Select the Subnet in which to create the endpoint.

Subnet
Select the subnets in which to create the endpoint.

subnet-0d66e2276cf7cfe6 (Private-Subnet-1)

IP address type

☒ IPv4

☐ IPv6

☐ Dualstack

Tags

Key Value - optional

Q Name X Q Private-Subnet-EC2-Endpoint X Remove

Add new tag

You can add 49 more tags.

Cancel Create endpoint

Wait until the endpoint status changes to **Available**. This indicates that the EC2 Instance Connect Endpoint has been successfully deployed and is ready for use.

Endpoints (1/1) [Info](#)

VPC endpoint ID : eice-0e3f37ab9abb37da2

Clear filters

Actions

Create endpoint

☒	Name	VPC endpoint ID	Endpoint type	Status	Service name
☒	Private-Subnet-EC2-En...	eice-0e3f37ab9abb37da2	EC2 Instance Connect Endpoint	Pending	-

Instances (1/2) Info

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

1

Settings

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IP
☐	Bastion-Host	i-0ab2cab66e77be4ef	Running	t3.micro	3/3 checks passed	us-east-1a	-
☒	Database-Server	i-0d5b9f5b0f4288f00	Running	t3.micro	3/3 checks passed	us-east-1a	-

EC2 Instance Connect

SSM Session Manager

SSH client

EC2 serial console

Instance ID

 i-Od5b9f5b0f4288f00 (Database-Server)

Connection type

☐ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☒ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Private IPv4 address
 192.168.0.182

☐ IPv6 address

EC2 Instance Connect Endpoint

Only endpoints that have completed the creation process can be selected. The process can take up to 15 minutes. If you create an endpoint, refresh this list to check if its in the available state.

ec2-0e3f37ab9abb37da2 (create-complete)

↻

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

✕

Max tunnel duration (seconds)

The maximum allowed duration of the SSH connection. Must comply with the maxTunnelDuration condition (if specified) in the IAM policy.

3600

Min 1 second. Max 3600 seconds (1 hour).

ⓘ

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

```

      _#_
     /##\   Amazon Linux 2023
    /###\
   /####\
  /#####\
 /#####\
/#####\
 \#####/
  \#####/
   \####/
    \###/
     \##/
      \|_
       V~' '-> https://aws.amazon.com/linux/amazon-linux-2023
         ^
        / \
       /   \
      /     \
     /       \
    /         \
   /           \
  /             \
 /               \
/                 \
 \               /
  \             /
   \          /
    \        /
     \      /
      \    /
       \  /
        \/
[ec2-user@ip-192-168-0-182 ~]$ █

```

Access Using a Bastion Host

Connect to the **Bastion Host** from your local computer using SSH and its **Public IP Address**. Since the Bastion Host resides in a public subnet, it is directly accessible from the internet.

[illegible]

Exit from the ec2-user to local system

exit

```

Last login: Tue Jun 30 08:48:41 2026 from 202.83.29.50
[ec2-user@ip-192-168-0-60 ~]$ exit
logout
Connection to 54.196.239.215 closed.

```

Copy your SSH private key (.pem file) from your local computer to the Bastion Host using the **scp** command. This allows the Bastion Host to authenticate when connecting to the Database Server.

```
scp -i <private-key.pem> <source-file> <username>@<destination-host>:<destination-path>
```

```
$ scp -i jenkins.pem jenkins.pem ec2-user@54.196.239.215:/home/ec2-user/  
jenkins.pem                                     100% 1678      6.3KB/s   00:00
```

```
amanr#LAPTOP-7OU8OST1 MINGW64 ~/Downloads  
$ |
```

After copying the key, connect to the Bastion Host and verify that the .pem file is present in your working directory. If necessary, modify the file permissions using `chmod 400 <key-name>.pem` so that SSH accepts the key.

[illegible]

From the Bastion Host, use SSH to connect to the **Database Server** by specifying its **Private IP Address**. Since both instances are inside the same VPC, the connection will succeed using the private network.

Open the **Routes** tab of the Main Route Table and click **Edit Routes**. Add a new route with the destination **0.0.0.0/0** and select the newly created **NAT Gateway** as the target, then save the route.

VPC > Route tables > rtb-039041c66222da988 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

Verify that the newly added route displays an **Active** status. This confirms that internet-bound traffic from the private subnets will now be forwarded to the NAT Gateway instead of attempting to reach the Internet Gateway directly.

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0c620c86ff74ad1b8	Active	No	Create Route
192.168.0.0/24	local	Active	No	Create Route Table

Connect back to the **Database Server** in the private subnet and test internet connectivity again by running commands such as `ping google.com`. The commands should now succeed, confirming that outbound internet access has been enabled.

```
[ec2-user@ip-192-168-0-182 ~]$ ping google.com
PING google.com (142.251.163.100) 56(84) bytes of data:
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=5 ttl=104 time=2.47 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=6 ttl=104 time=2.04 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=7 ttl=104 time=2.02 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=8 ttl=104 time=2.04 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=9 ttl=104 time=2.04 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=10 ttl=104 time=2.04 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=11 ttl=104 time=2.00 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=12 ttl=104 time=2.11 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=13 ttl=104 time=2.03 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=14 ttl=104 time=2.01 ms
64 bytes from vv-in-f100.1e100.net (142.251.163.100): icmp_seq=15 ttl=104 time=2.14 ms
^C
--- google.com ping statistics ---
15 packets transmitted, 11 received, 26.6667% packet loss, time 14195ms
rtt min/avg/max/mdev = 2.002/2.083/2.469/0.128 ms
[ec2-user@ip-192-168-0-182 ~]$
```

Note: **NAT Gateway is a paid AWS service**. After completing the lab, delete the NAT Gateway to avoid unnecessary charges.